

Flutter & Git Basics

Git version control · Widget architecture · Stateless vs stateful · UI layout

Learning Goals

- Use Git confidently in Flutter projects and push to GitHub
- Understand the Flutter widget tree and build method
- Differentiate StatelessWidget from StatefulWidget
- Build basic UI layouts with core and layout widgets
- Create and navigate between multiple Flutter screens

1. Git for Flutter Developers

Daily Workflow

Check what changed

git status

Stage everything

git add .

Save a snapshot

git commit -m "Add profile card UI"

Sync with GitHub

git push origin main

git pull origin main

Branching

git branch feature-ui # create branch

git checkout feature-ui # switch to it

... do work, commit ...

git checkout main

git merge feature-ui # merge back

Good Commit Messages

Use the imperative mood. Be specific. Keep it under 72 characters.

- Add profile card UI
- Implement stateful counter screen
- Fix navigation issue on Android
- Refactor home screen layout

Flutter .gitignore

```
.dart_tool/  
.packages  
build/  
.idea/  
android/.gradle/  
ios/Pods/  
.env
```

First Push to GitHub

```
git init  
git add .  
git commit -m "Initial Flutter project setup"  
git remote add origin https://github.com/you/project.git  
git branch -M main  
git push -u origin main
```

2. Flutter Architecture

Everything in Flutter is a widget — text, buttons, entire screens, even the app itself.

Widget Tree

```
MaterialApp  
├── Scaffold  
│   ├── AppBar  
│   │   └── Text('My App')  
│   ├── Center  
│   │   └── Text('Hello, Flutter!')
```

Hot Reload vs Hot Restart

r · hot reload — Repaints UI, keeps state.

R · hot restart — Full restart, clears state.

Recommended Folder Structure

```
lib/  
├── main.dart  
├── screens/  
│   ├── home_screen.dart  
│   └── profile_screen.dart  
├── widgets/  
├── models/  
└── utils/
```

3. Stateless vs Stateful Widgets

StatelessWidget — No Mutable State

```
class WelcomeText extends StatelessWidget {  
  final String name;  
  const WelcomeText({required this.name});  
  
  @override  
  Widget build(BuildContext context) {  
    return Text('Welcome, $name!');  
  }  
}
```

StatefulWidget — Reacts to Changes

```
class CounterScreen extends StatefulWidget {  
  @override  
  State<CounterScreen> createState() => _CounterScreenState();  
}  
  
class _CounterScreenState extends State<CounterScreen> {  
  int _count = 0;  
  
  void _increment() => setState(() => _count++);  
  
  @override  
  Widget build(BuildContext context) {  
    return Column(children: [  
      Text('Count: $_count'),  
      ElevatedButton(onPressed: _increment, child: Text('+')),  
    ]);  
  }  
}
```

Always wrap state mutations inside `setState()`. Without it, Flutter won't redraw the UI.

BuildContext & Navigation

```
Navigator.push(  
  context,  
  MaterialPageRoute(builder: (_) => SecondScreen()),  
);
```

4. Core Widget Reference

Widget	Purpose
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MaterialApp	Root widget — provides theme, routes, title
Scaffold	Page template: AppBar, body, FAB, drawer
AppBar	Top bar with title and action buttons
Text / RichText	Render text. RichText supports mixed styles.
Container	Box with width, height, padding, decoration
Row / Column	Horizontal / vertical children layout
Stack	Layer children on top of each other
Padding / Center	Add space around a child / center it
SizedBox	Fixed-size box; also used as a spacer
Expanded / Flexible	Distribute remaining space in Row/Column
Spacer	Fills available space — pushes siblings apart
ListView	Scrollable list of widgets

5. UI Practice Projects

- Profile card — CircleAvatar, Column, Card inside StatelessWidget
- Simple form — TextField, validators, ElevatedButton, StatefulWidget
- List screen — ListView.builder rendering a Dart List
- Bottom nav bar — Scaffold.bottomNavigationBar + IndexedStack

6. Resources

YouTube Channels

- Flutter Official
- The Flutter Way
- Flutter Mapp
- Reso Coder
- The Net Ninja

Search Terms

- Git for Flutter developers
- Flutter widgets explained
- Stateless vs stateful
- Flutter UI basics

- Flutter MaterialApp tutorial

Deliverables Checklist

1. Git repository with 5+ meaningful commits
2. Flutter app using 8+ different widgets
3. At least one StatelessWidget and one StatefulWidget with real state
4. 3+ UI screens (profile card, form, list, nav bar)